

Wen Fan

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RESEARCH SUMMARY

Ph.D. researcher in software verification and static analysis. I combine separation-logic-based verification and LLM-assisted specification generation for heap-manipulating programs, aiming to apply formal methods to security-critical and distributed systems.

EDUCATION

Ph.D., Electrical and Computer Engineering, Purdue University

West Lafayette, Indiana, USA

- Advisor: Prof. Jenna DiVincenzo

Aug. 2022 – Aug. 2028 (*Expected*)

B.S., Computer Science, University of Science and Technology of China (USTC)

Hefei, Anhui, China

- GPA: 3.75/4.3

Sept. 2018 – Jun. 2022

PUBLICATIONS

- **Wen Fan**, Minh Tran, Sanya Dod, Xin Hu, Marilyn Rego, Danning Xie, Jenna DiVincenzo, Lin Tan
An Empirical Study of LLM-Generated Specifications for VeriFast
The 37th IEEE International Symposium on Software Reliability Engineering (**ISSRE 2026**)
- Marilyn Rego*, **Wen Fan***, Xin Hu, Sanya Dod, Zhaorui Ni, Danning Xie, Jenna DiVincenzo, Lin Tan
Evaluating the Ability of Large Language Models to Generate Verifiable Specifications in VeriFast
The 2nd ACM international conference on AI Foundation Models and Software Engineering (**FORGE 2025**) [[paper](#)][[code](#)]

RESEARCH PROJECTS

Evaluating and Enhancing LLMs on Generating Specifications for Verifying Heap-manipulating programs Purdue University

Advisor: Prof. Jenna DiVincenzo

Sept. 2024 – Now

- Built a benchmark based on 60 programs specified in VeriFast verifier
- Developed automated Python pipeline to evaluate 10 LLMs on 2700+ outputs
- Led our team of 4 to inspect and categorize 1648 VeriFast errors from 450 sampled output functions
- Identified only 31.7% verification success; Over 87% functional behavior preserved; 71% of errors caused by tool-specific knowledge gaps (e.g., missing open/close); 47% of errors can be fixed by modifying simple auxiliary specifications
- Building the tool to improve the performance based on the empirical findings

Optimizing Gradual Program Verifier with Parallel Symbolic Execution

Purdue University

Advisor: Prof. Jenna DiVincenzo

Jun. – Aug. 2024

- Discovered the performance bottleneck in sequential execution by analyzing a program of 571 branches
- Designed and implemented parallel execution of branches, reducing static verification time by 41%

Consensus Protocol Optimization under Different Fault Models

Purdue University

Advisor: Prof. Yongle Zhang

May. 2023 – Mar. 2024

- Investigated 40+ consensus protocols in different fault models (e.g., ordered network, reduced node failures)
- Conducted a survey about 5+ synthesis techniques on distributed systems/protocols with 3 other team members
- Identified 8+ examples of improving the performance of Paxos under different fault models

Static Analysis For Failure Detection in Distributed Systems

Purdue University

Advisor: Prof. Yongle Zhang

Jun. 2022 – Apr. 2023

- Examined 100+ failure cases on Apache Jira to characterize cascading failures
- Implemented a static analyzer using the Soot framework to recognize 7 cases of retry in HBase

Optimizing static taint analyzer

Remote research intern at UIUC

Advisor: Prof. Tianyin Xu

Jul. – Oct. 2021

- Added flow information for each statement to solve the non-deterministic path reconstruction
- Implemented points-to analysis and field-use analysis to handle false positives

Index Optimization on Key-Value Storage

Research intern at USTC

Advisor: Prof. Yongkun Li

Sept. 2020 – Dec. 2021

- Co-designed and implemented the bloom filter and iterator in the LSM tree for a new indexing in LevelDB source code
- Used perf in Linux to evaluate the performance of the new indexing

COURSE PROJECTS

DSLAB (Sharded and Fault Tolerant Distributed Key-Value Store)

CS505 course at Purdue University

- Implemented Primary-Backup, Paxos and Two Phase Commit protocols
- Passed more than 90% of tests by debugging on searching tests and corner cases

Tiny MIPS Pipelined CPU

Computer Organization and Design course at USTC

- Built a 5-stage pipelined CPU of simplified MIPS instruction set, validated by simulation and FPGA testing

OSLAB

Operating System course at USTC

- Simulated shell (by using pipes), malloc and a FAT file system (by using fuse)

Compiler Lab

Compiler Principle course at USTC

- Implemented lexical analysis, parsing and intermediate representation generation for C- (a subset of C)

Simple Bank Management System

Database course at USTC

- Implemented the CRUD features in a bank system using Django, where the E-R models are designed based on the user's requirement

TALKS

- Evaluating the Ability of Large Language Models to Generate Verifiable Specifications in VeriFast, FORGE @ ICSE 2025
- Optimizing Gradual C0 Program Verifier by Adding Parallelism, Purdue ECE Symposium 2024

MENTORING

I am lucky to work with Marilyn Rego, Sanya Dod and Minh Tran who are (were) excellent undergraduates at Purdue University.

TEACHING ASSISTANT EXPERIENCE

- Mathematical Logic USTC, Spring 2021
- CS252 Systems Programming Purdue University, Fall 2022, Spring 2023, Spring 2024
- CS251 Data Structures Purdue University, Fall 2023

AWARDS

Huawei Scholarship Nov. 2020
Huaxia Talent Program, USTC Jun. 2022

SKILLS

Programming Languages: C, C++, Python, Java, Scala, Go, Rust, OCaml, Verilog, SQL, MATLAB

Verification/Analysis: VeriFast, Dafny, Verus, Rocq, Soot, LLVM

Systems/Tools: Linux, Git, Docker, Makefile, LaTeX, Django, Wireshark

Languages: English (fluent), Chinese (native)